

Mon		Wed		Fri		Sun
Tue		Thu		Sat		

Date: 21-08-23

Book $\Rightarrow$ Microelectronics circuit 6th Edition, International version.

Resistor
Capacitor
Inductance

Linear because

$$V = IR$$

$$Q = CV$$

$$\phi = Li$$

It is the part of DC supply.

Diode

Transistor

non-linear

BJT

Bipolar Junction
Transistor

MOSFET

(Metaloxide Semiconductor Field
Effect Transistor)

Rectifier Diode
Zener Diode

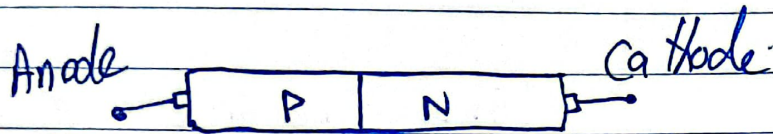
Diode can use as a switches.

Transistor can be use as

$\Rightarrow$ amplifier
 $\rightarrow$ Switch

The P-N Junction

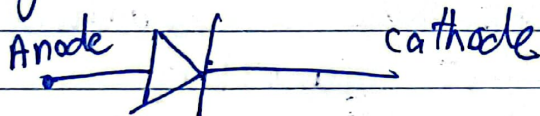
∴ The process of adding impurity to conductor is called doping



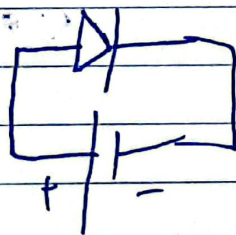
⇒ N-Type semiconductor
more -ve
⇒ Pentavalent

P-Type semiconductor
more +ve
trivalent

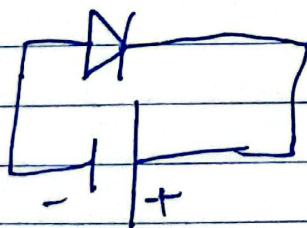
Symbol



DC sources
AC sources



forward Biasing



Reverse Biasing

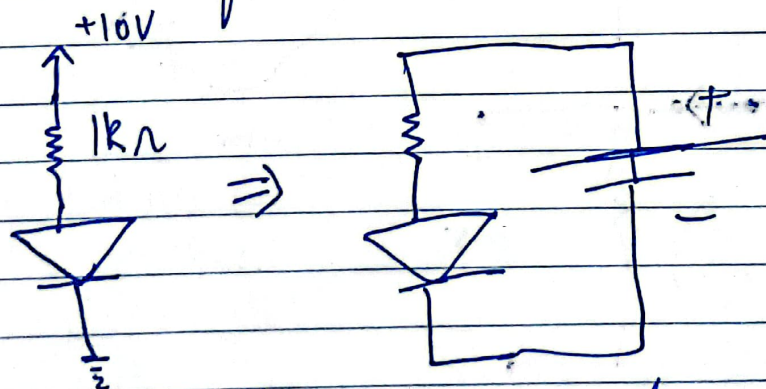
The ideal diode:-

$\Rightarrow$ If diode is forward biasing behave as close switch or $\cup$ a short circuit

$\Rightarrow$ If diode is reversed biasing behave as open circuit or $\cup$ a no current pass through the circuit.

If diode is conducting it should be forward biasing.

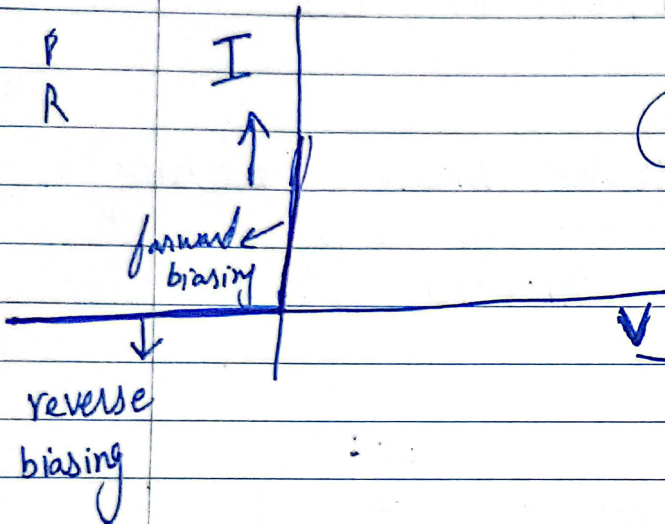
Condition for conduction



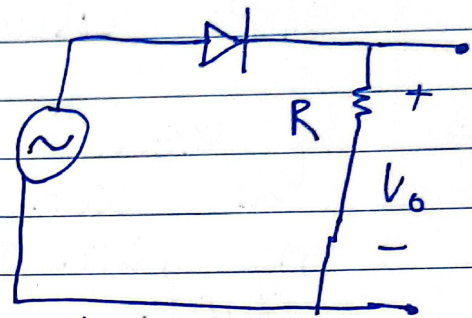
for forward biasing Anode potential is higher than the potential at cathode and vice versa

V-I characteristic of the ideal diode.

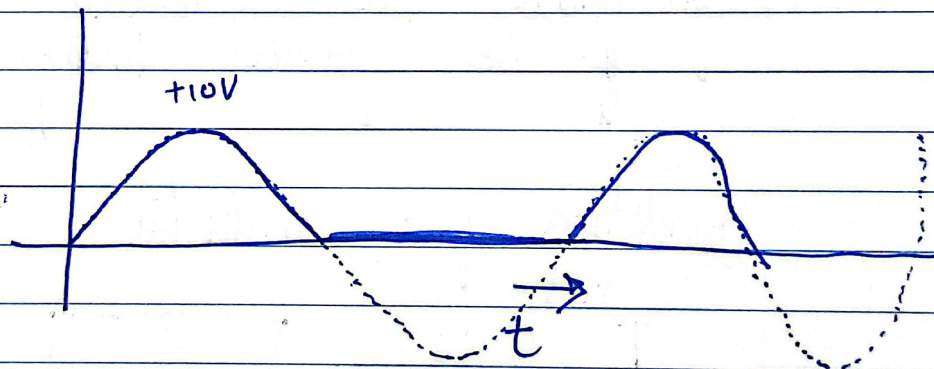
Diode in AC circuit:-



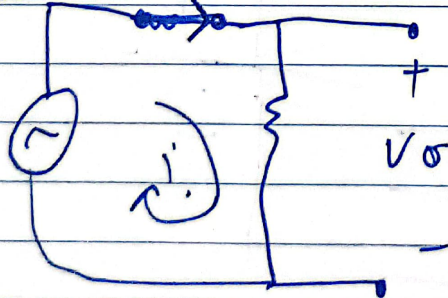
①



Sketch output voltage



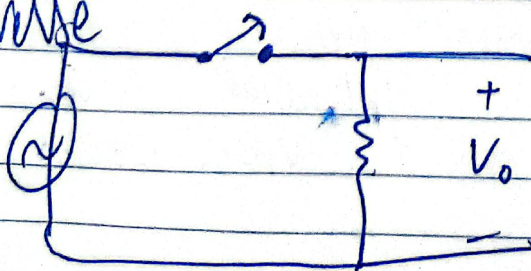
Forward



-10V

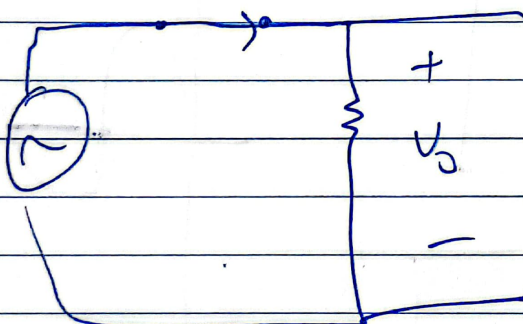
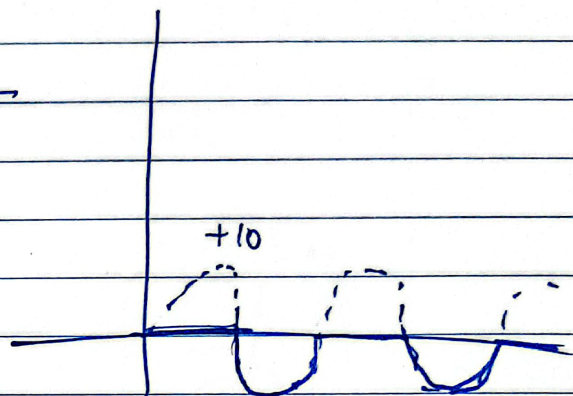
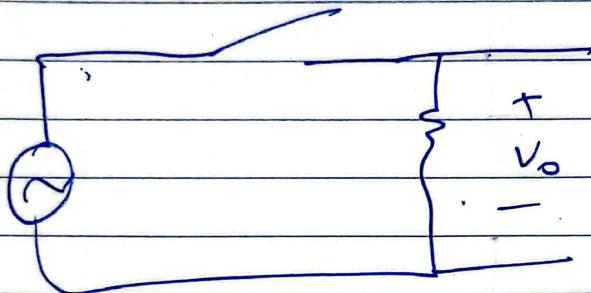
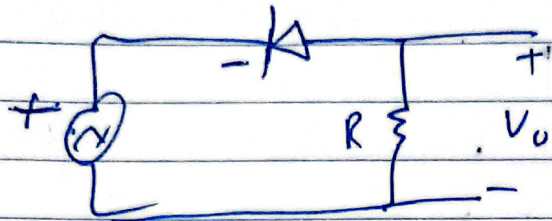
⇒ output follows the input in positive half cycle

Reverse

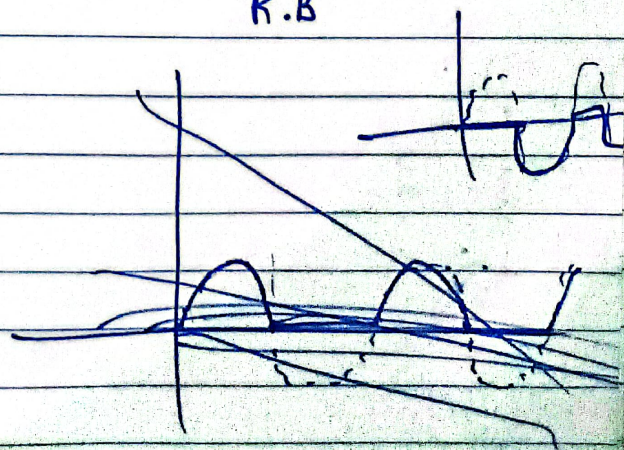
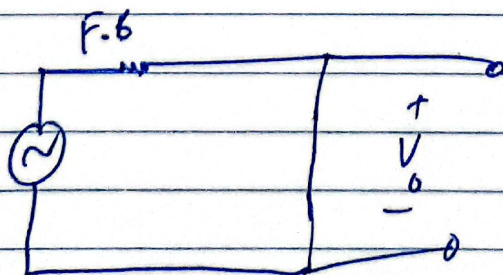
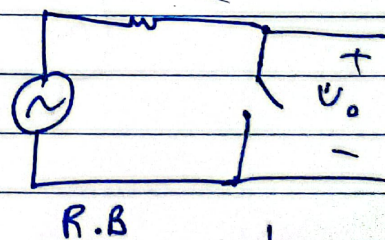
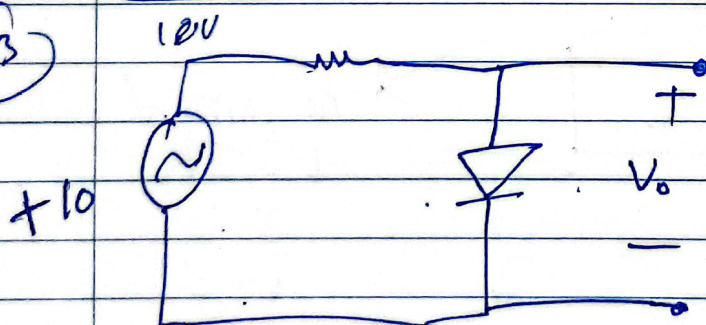


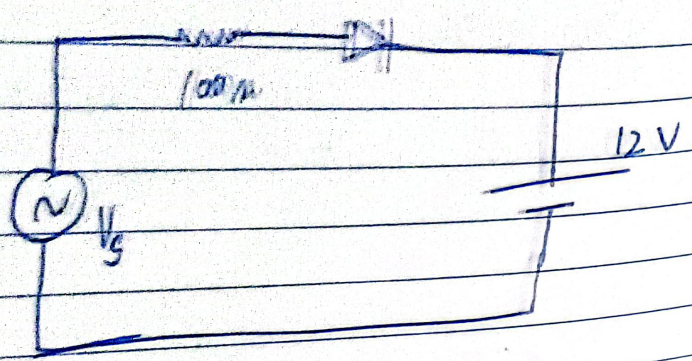
⇒ output is zero in second half cycle.

2



3

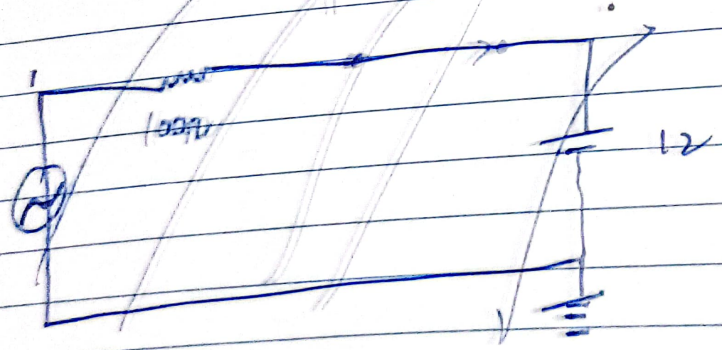
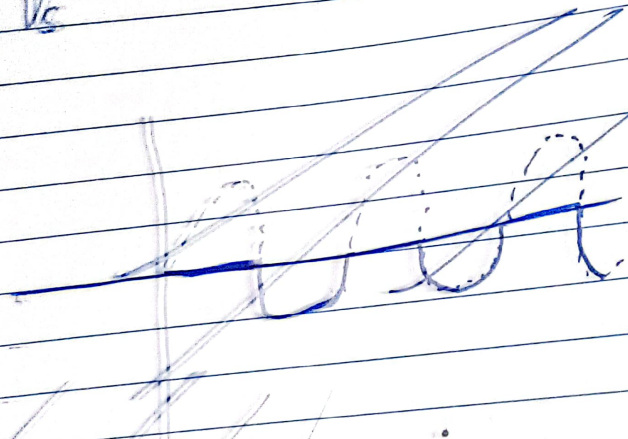


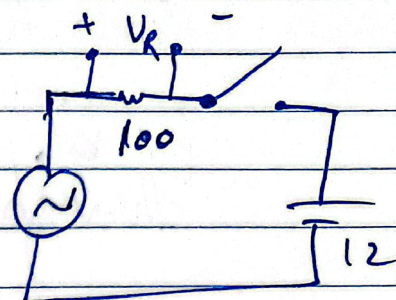
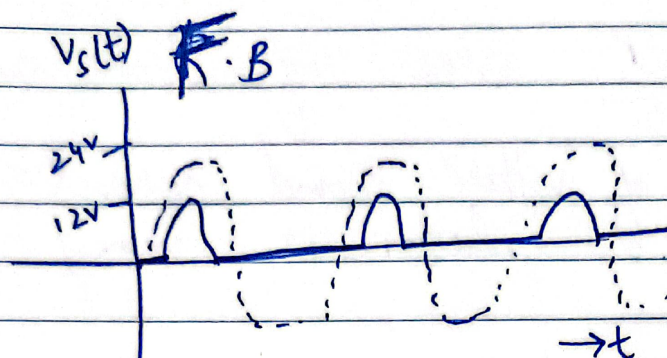


V_s is a sinusoidal of 24 V peak

(a) sketch V_s

F.B





(b) Find the fraction of the cycle during which diode is conduction angle $= \theta_1, \theta_2$ conducts

$$\theta_1 = ?$$

$$\theta_2 = ?$$

$$24 \sin \theta_1 = 12$$

$$\sin \theta = \frac{1}{2}$$

$$\theta_2 = \sin^{-1}\left(\frac{1}{2}\right)$$

$$\theta_1 = 30^\circ$$

$$\theta_2 = 180^\circ - 30^\circ = 150^\circ$$

$$\text{conduction angle} = 150^\circ - 30^\circ = 120^\circ$$

$$\text{fraction} = \frac{120}{360}$$

$$\text{fraction} = \frac{1}{3} = 33.33\%$$

c) Find the peak value of diode current.
 (∵ ^{waveform} shape of voltage and current flowing through the resistor is same)

$$V = IR$$

$$I = \frac{V}{R}$$

$$= \frac{12}{100}$$

$$I = 0.12 \text{ amp}$$

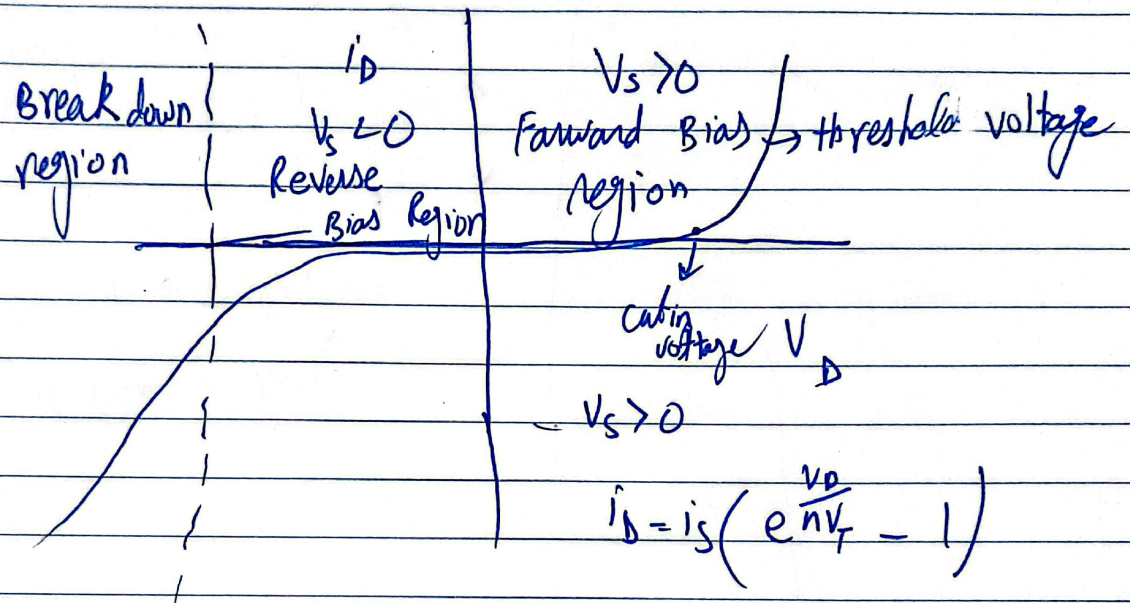
Peak voltage = 12 $\Rightarrow$ (24 - 12) ^{Peak AC} $\rightarrow$ DC

d) Find the ^{magnitude} maximum voltage that appears across diode in reverse bias?

Peak Inverse Voltage:- PIV (data sheet parameter)

$$\begin{aligned} PIV &= V_A - V_C \\ &= (-24 - 12) \text{ V} \\ &= -36 \text{ V} \end{aligned}$$

IV characteristics of Real/Practical diode



i_S = saturation / scale current

- for small diode of the order of $10^{-15} A$
- depends on junction area of diode
- doubles for every $5^\circ C$ rise in temperature
- doubles by doubling Junction area.

V_T = thermal voltage = 25 mV

$$= \frac{kT}{q}$$

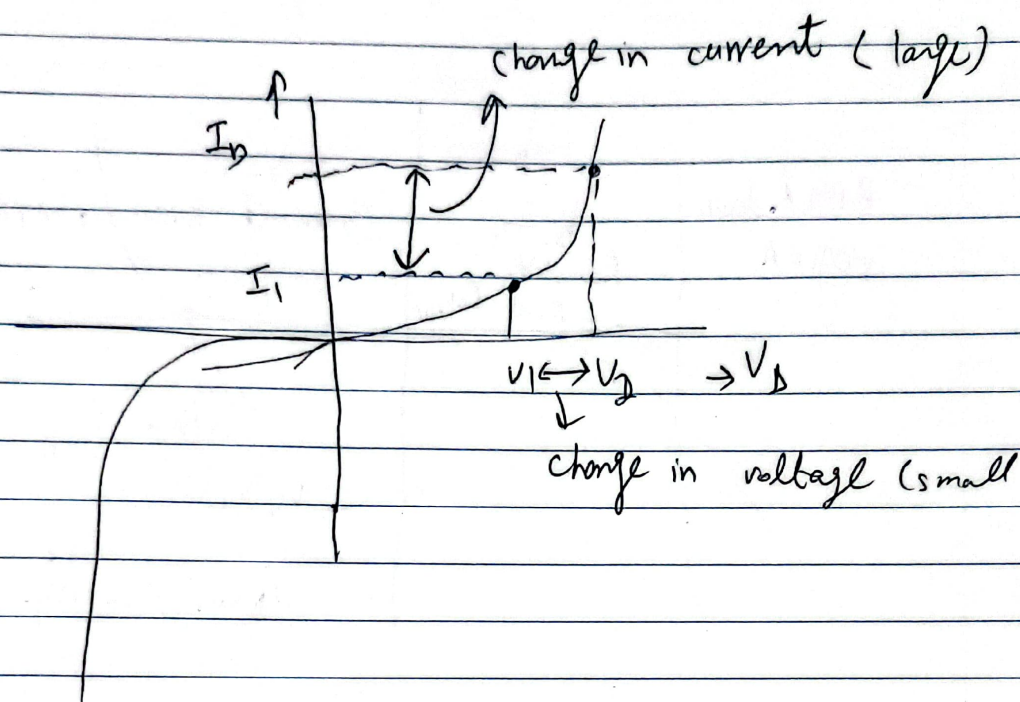
k = Boltzmann's constant ($1.38 \times 10^{-23} J/K$)

T = temperature in kelvin

q = charge on electron ($1.6 \times 10^{-19} C$)

$$n = 1 \text{ to } 2$$

$$I_D = I_S \left(e^{\frac{V_D}{nV_T}} - 1 \right)$$



$$I_D = I_S e^{\frac{V_D}{nV_T}} - I_S \quad (\text{for our case})$$

$$I_D \approx I_S e^{\frac{V_D}{nV_T}}$$

$$I_2 = I_S e^{\frac{V_2}{nV_T}} \quad \text{--- (1)}$$

$$I_1 = I_S e^{\frac{V_1}{nV_T}} \quad \text{--- (2)}$$

Divide (1) & (2)

$$\frac{I_2}{I_1} = \frac{I_S e^{\frac{V_2}{nV_T}}}{I_S e^{\frac{V_1}{nV_T}}}$$

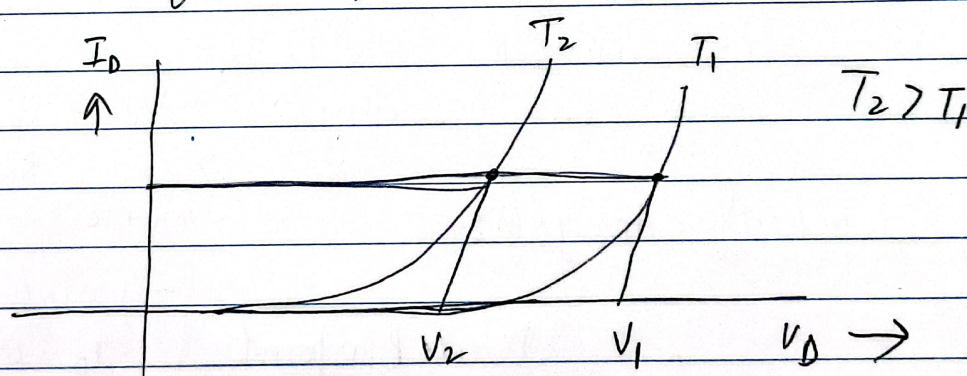
$$\frac{I_2}{I_1} = e^{\frac{V_2 - V_1}{nV_T}}$$

$$\Rightarrow V_2 - V_1 = n V_T \ln \frac{I_2}{I_1}$$

$$\Rightarrow V_2 - V_1 = 2.3 V_T \log_{10} \frac{I_2}{I_1}$$

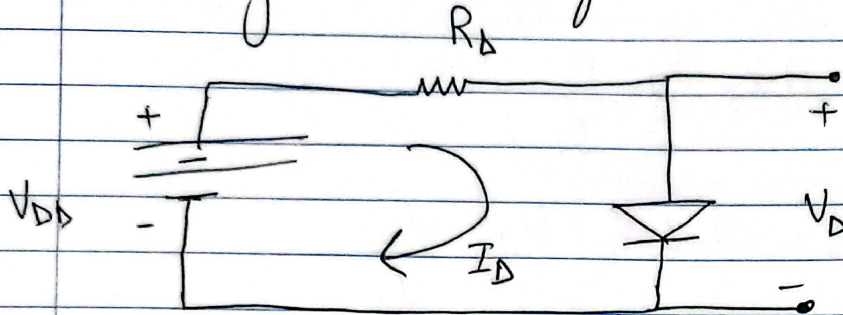
(small) linear change in voltage corresponds to logarithmic (large) change in current.

$\Rightarrow$ Effect of temperature on diode



$\Rightarrow V_D$ decreases by 2mV per degree rise in temperature.

Modeling the diode forward characteristics



Analyze the circuit to find V_D & I_D

① The Exponential Model:-

$$I_D = I_S e^{V_D / nV_T} \quad \text{--- (1)}$$

$$I_D = \frac{V_{DD} - V_D}{R_D} \quad \text{--- (2)}$$

$$\Rightarrow \text{re arrange } \Rightarrow \frac{-V_D}{R_D} + \frac{V_{DD}}{R_D}$$

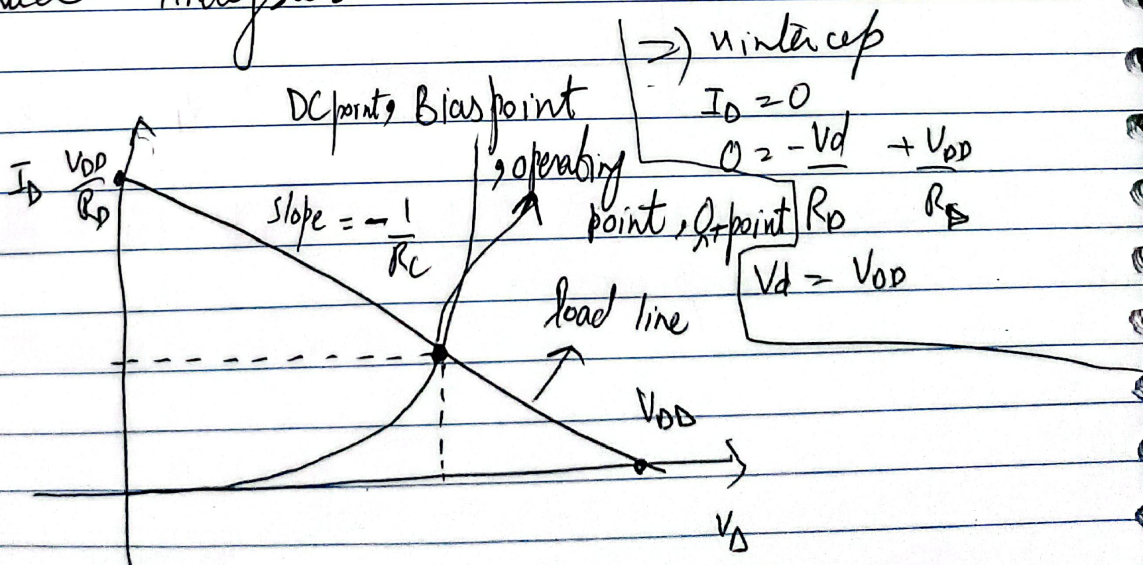
$\uparrow$
 $mx + c$

Graphical Analysis:-

Y-intercept

$$V_D = 0$$

$$I_D = \frac{V_{DD}}{R_D}$$



Iterative Analysis :-

Determine V_D and I_D $V_{DD} = 5V, R_D = 1k\Omega$
 assume that a diode has current 1 mA at the voltage drop across $0.7V$ and its voltage drop changes by 0.1 volt for every decade (10) change in current.

Sol.:-

$$V_2 - V_1 = 2.3 n V_T \log \frac{I_2}{I_1}$$

for decade current

$$V_2 - V_1 = 2.3 n V_T \log_{10} \frac{10 I_1}{I_1}$$

$$V_2 - V_1 = 2.3 n V_T$$

$$V_2 - V_1 = 0.1V$$

Assume $V_D = 0.7V$

$$I_D = \frac{V_{DD} - V_D}{R_D}$$

$$I_D = \frac{5 - 0.7}{1k}$$

$$I_D = 4.3 \text{ mA}$$

$$I_1 = 1 \text{ mA} \quad V_1 = 0.7V$$

$$I_2 = 4.3 \text{ mA} \quad V_2 = ?$$

$$V_2 - V_1 = 2.3 n V_T \log \frac{I_2}{I_1}$$

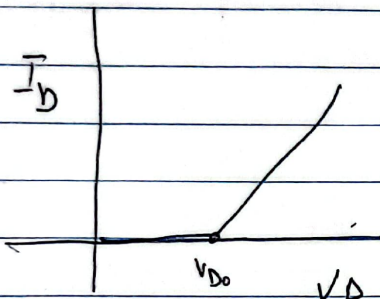
$$V_2 - 0.7 = (0.1) \log \frac{4.3 \text{ mA}}{1 \text{ mA}}$$

$$V_2 = 0.7 + (0.1) \log 4.3 \text{ mA}$$

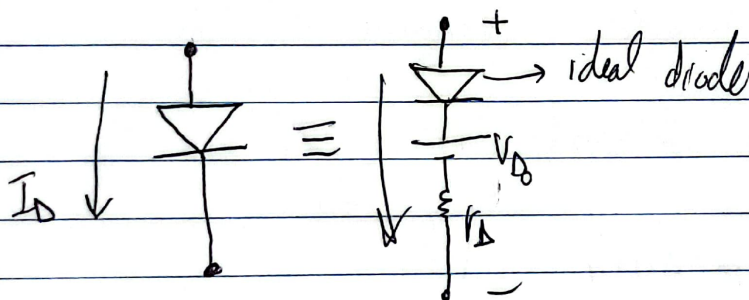
$$\boxed{V_2 = 0.763 \text{ V}}$$

Keep on doing the iterations until there is no change in current.

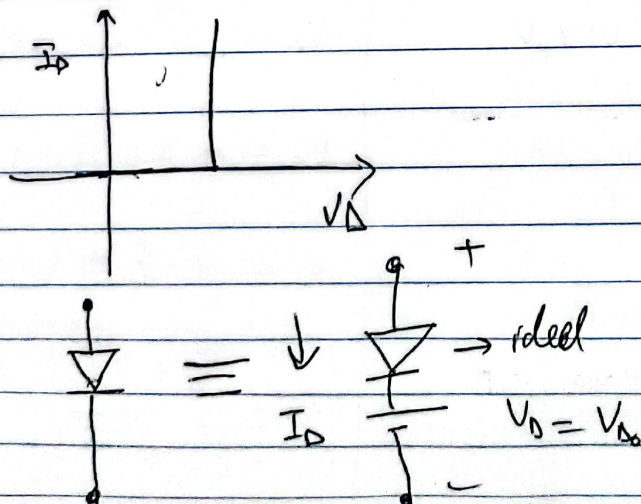
Piecewise Linear Model



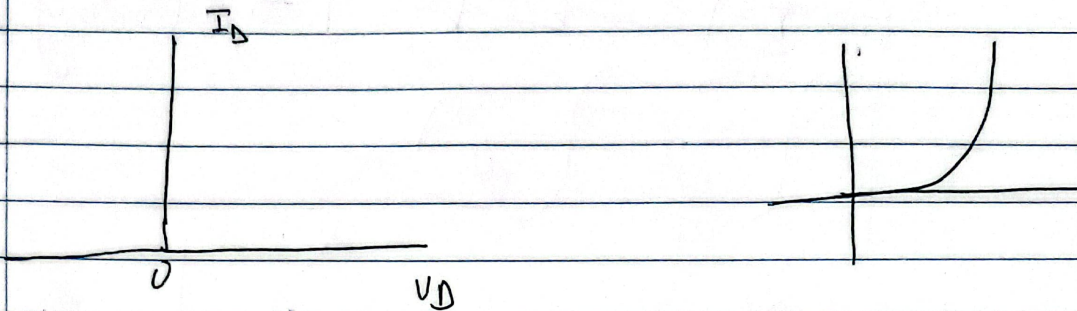
$$V_D = V_{D0} + I_D r_D$$



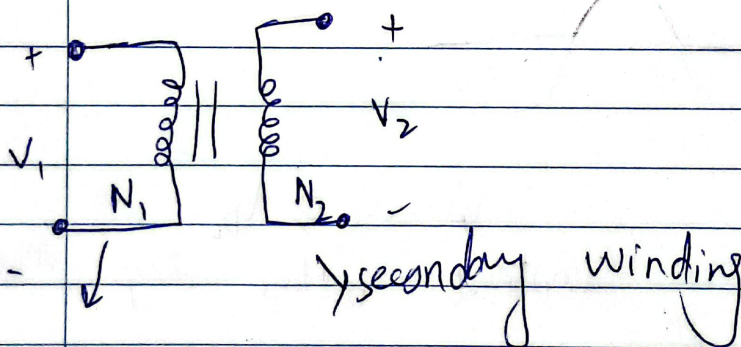
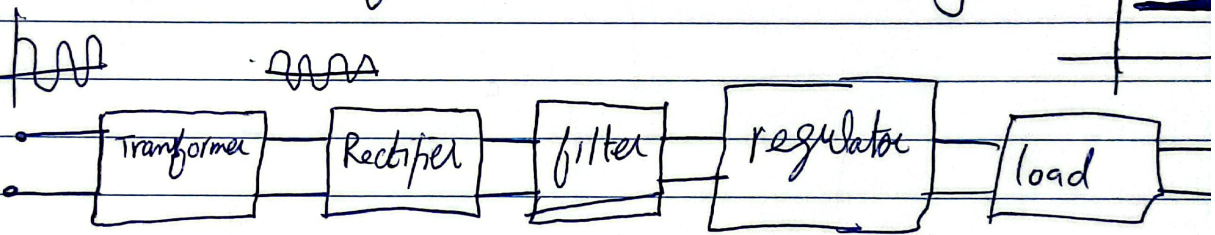
constant Voltage Drop Model:-



Ideal Diode Mode



Components of DC Power Supply:-



$$\frac{V_2}{V_1} = \frac{N_2}{N_1}$$

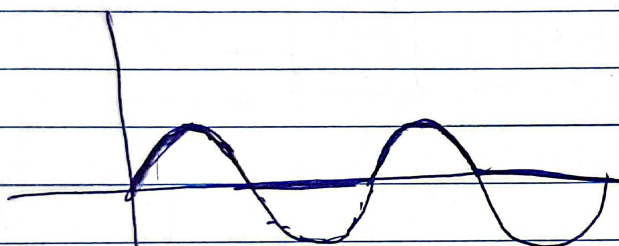
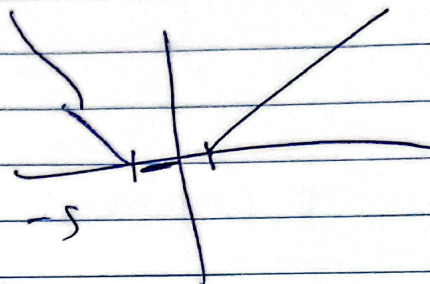
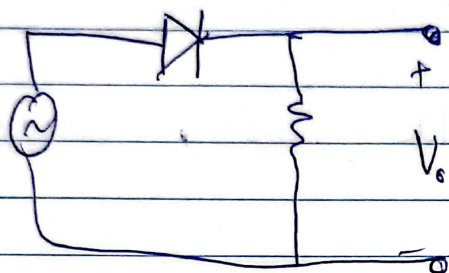
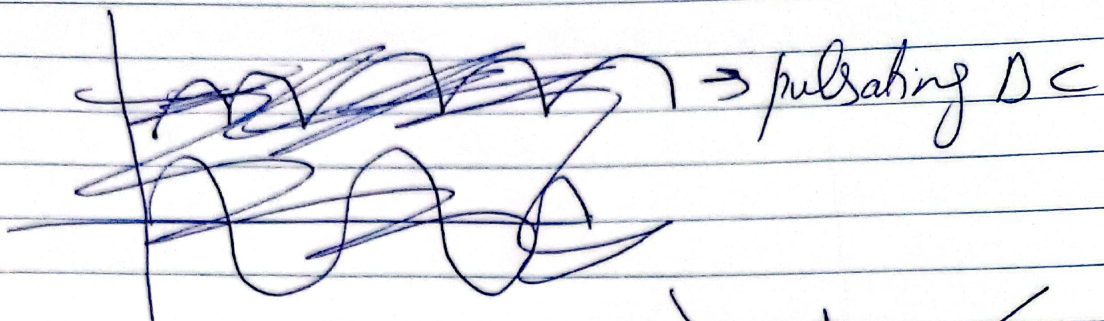
Primary winding

Rectifier:-

Rectifier is a device which convert AC into pulsating DC

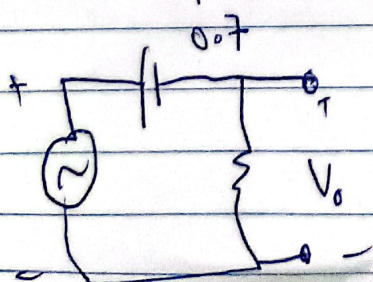
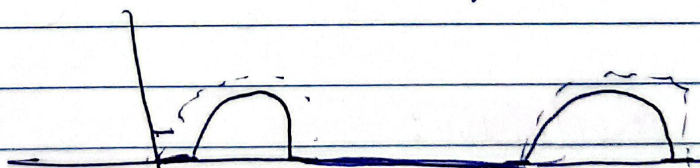
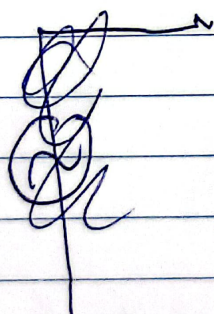
Type:-

- Half wave Rectifier
- Full wave Rectifier
- Center tapped Fw
- Bridge Rectifier



half wave rectifier

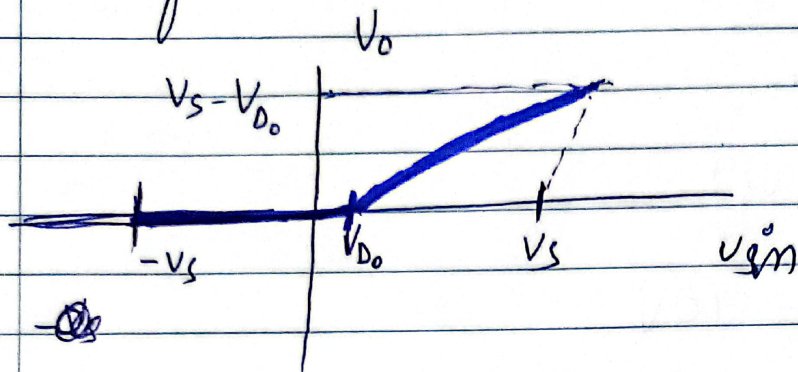
$V_A - V_C \geq V_{D0}$
constant voltage drop Model: -



$$V_o = V_{in} - V_D$$

$$V_o = 9.3$$

Transfer characteristics:-



Non-zero Average (Piecewise)

$$V_{avg} = \frac{1}{T} \int u(t) dt$$

$$= \frac{1}{2\pi} \int_0^{\pi} V_S \sin \phi d\phi + \int_{\pi}^{2\pi} 0 d\phi$$

$$= \frac{V_S}{2\pi} [-\cos \phi]_0^{\pi}$$

$$= \frac{V_S}{2\pi} [\cos(\pi) - \cos(0)]$$

$$= \frac{V_S}{2\pi} [-1 - 1]$$

$$= \frac{V_S}{\pi} \Rightarrow 0.318 V_S$$

PIV :-

$$V_C - V_A$$

$$= 0 - (-V_s)$$

$$= V_s = 10V$$

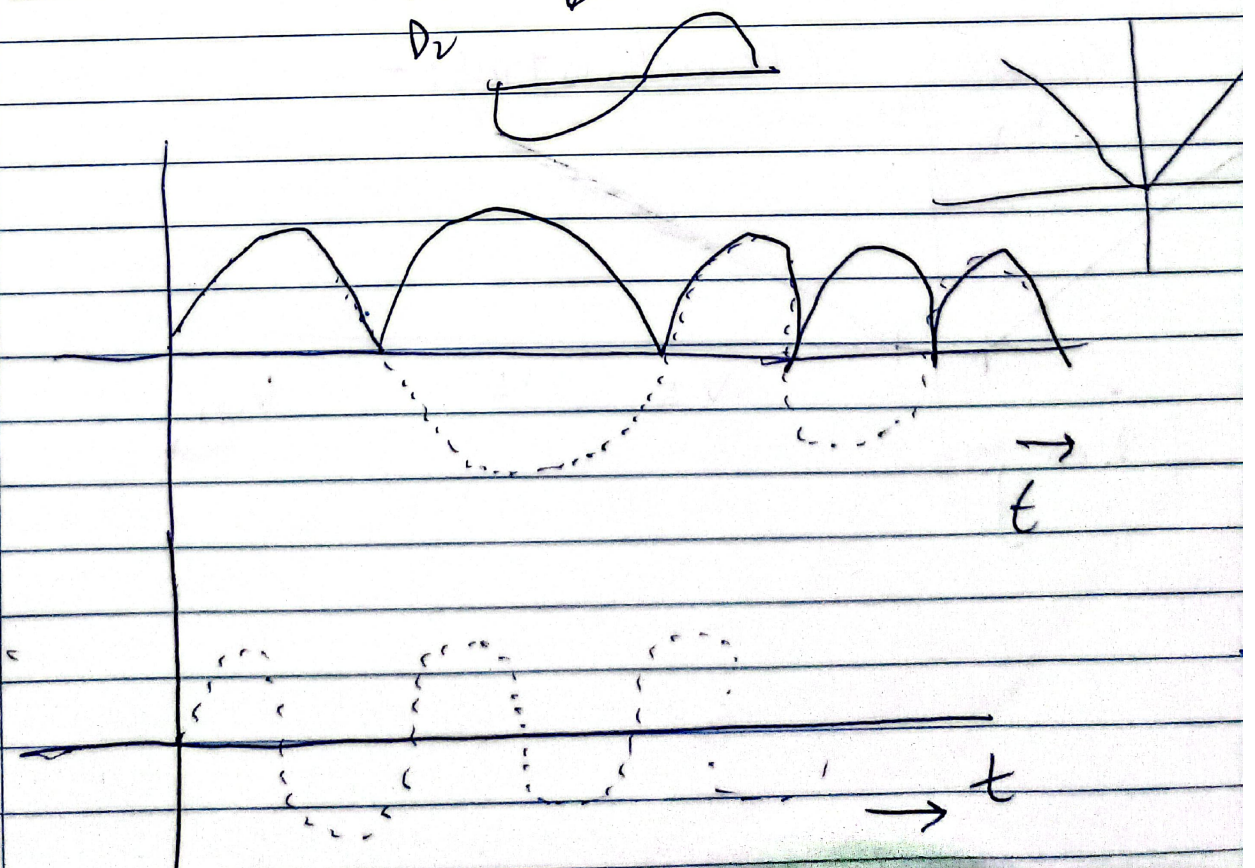
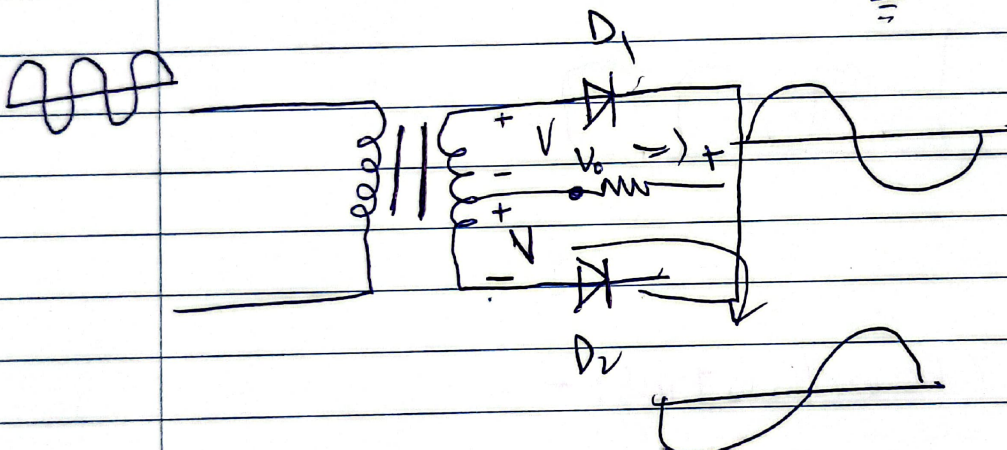
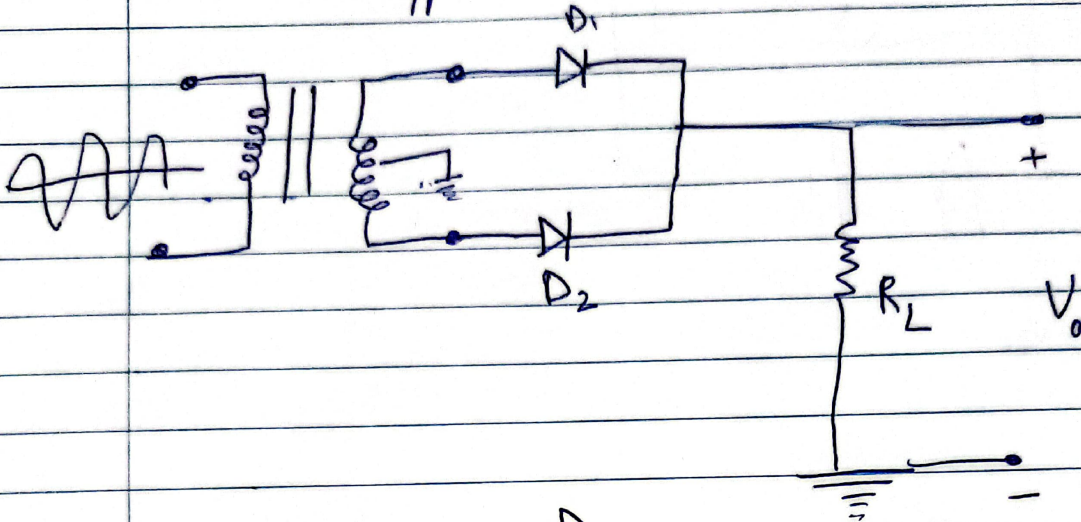
Controlled Model (Average value)

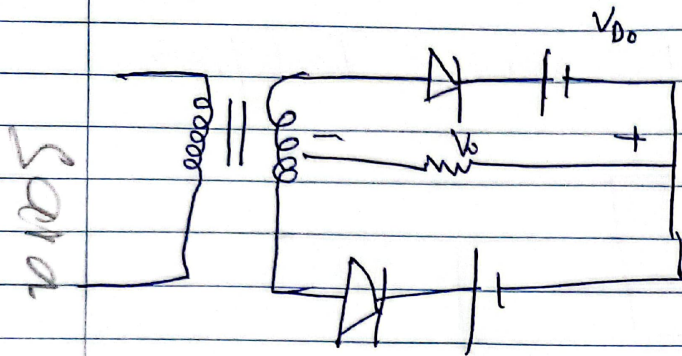
$$V_{avg} = \frac{1}{2\pi} \left[\int_{\pi-\theta}^{\theta} 0 d\phi + \int_{\theta}^{\pi-\theta} (V_s \sin\phi - V_{D0}) d\phi + \int_{\pi-\theta}^{2\pi} 0 d\phi \right]$$

$$V_{avg} = \frac{1}{2\pi} \left[\int_{\pi-\theta}^{\theta} (V_s \sin\phi - V_{D0}) d\phi \right]$$

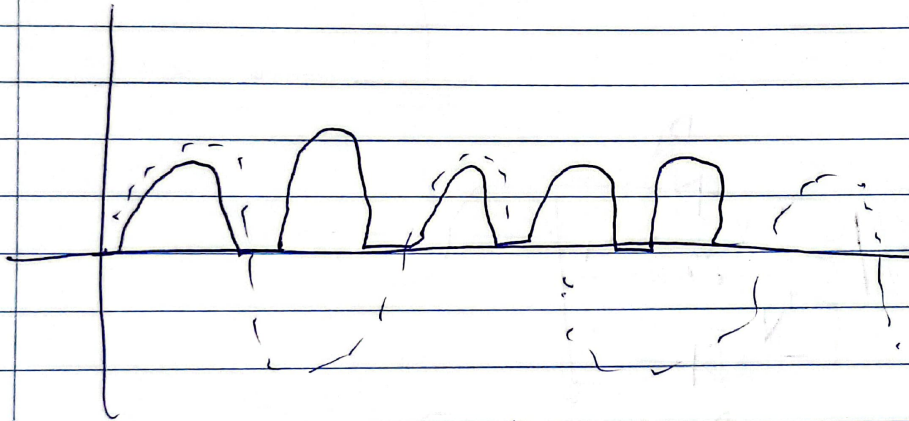
$$= \frac{V_s}{2\pi} \cos\theta - \frac{V_{D0}}{2\pi} [\pi - 2\theta]$$

Center Tapped Full wave Rectifier:-

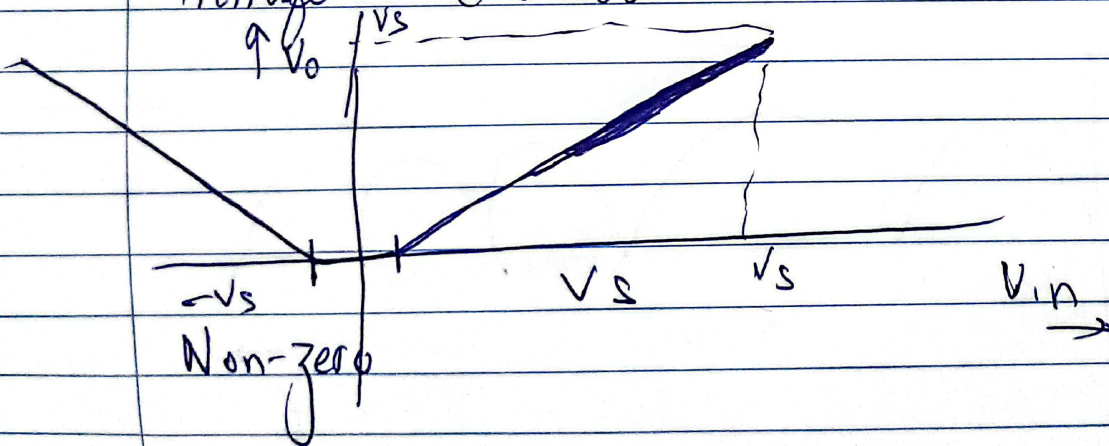


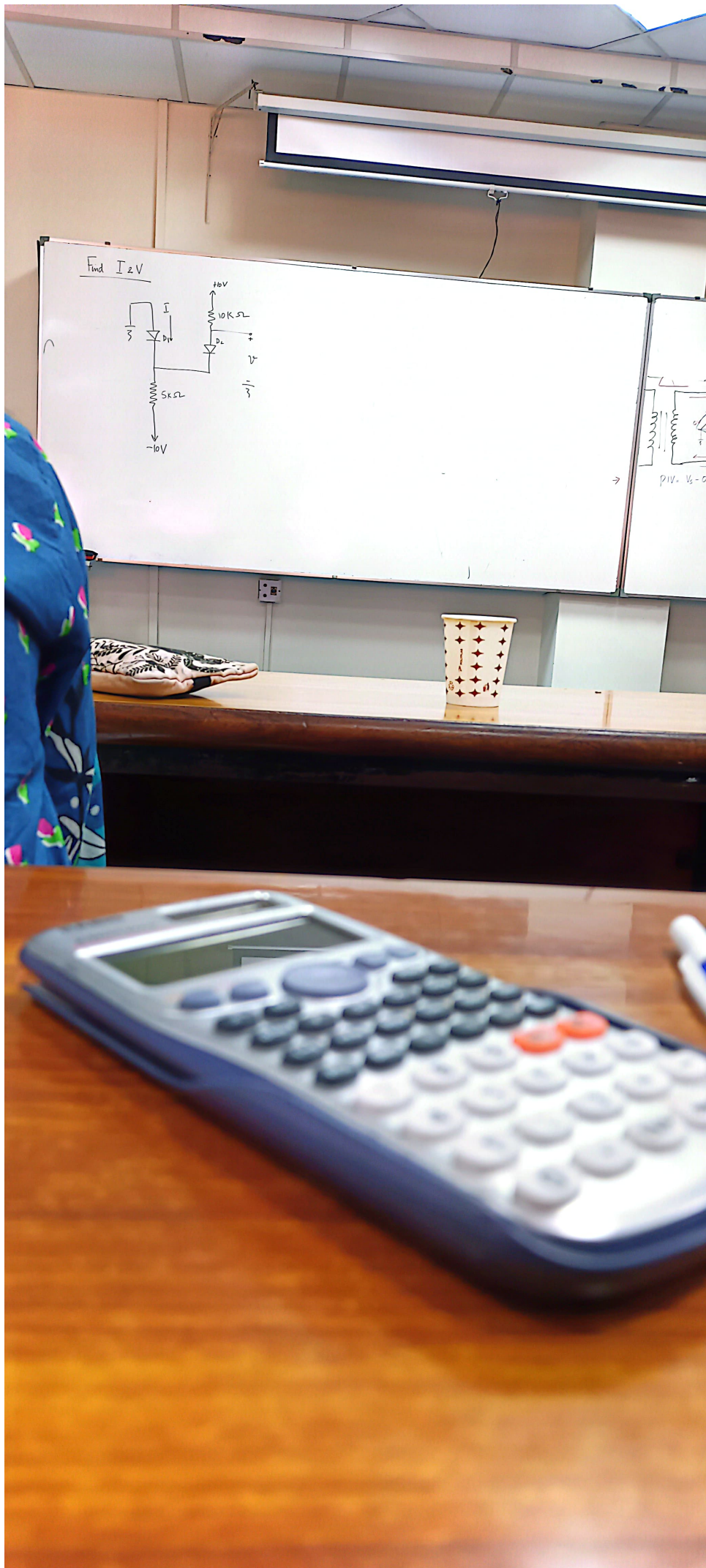


$$V_s - V_{D0}$$

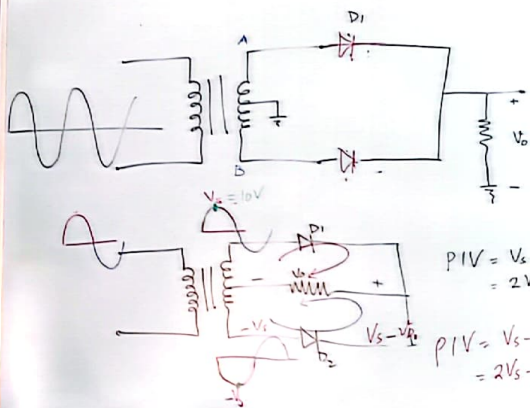


Transfer Characteristics:-



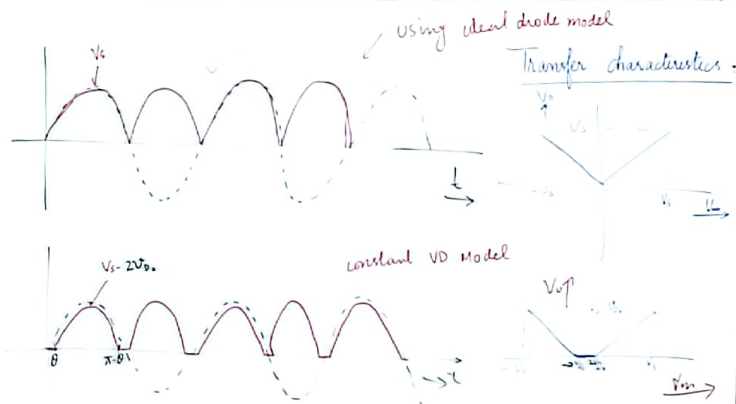


Center Tapped Fullwave Rectifier:

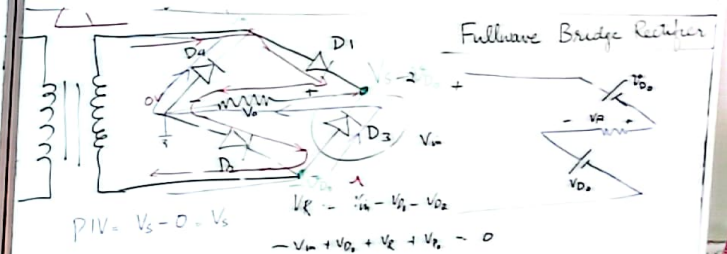


$$PIV = V_s - (-V_s) = 2V_s$$

$$PIV = V_s - V_{s1} - (-V_s) = 2V_s - V_{s1}$$



$$PIV = (V_s - 2V_{D0}) - (-V_{D0}) = V_s - V_{D0}$$



$$PIV = V_s - 0 = V_s$$

$$-V_{in} + V_{D0} + V_L + V_{D0} = 0$$